

A Zero-Knowledge Proof of Handlebody Spine Embedding via Integer Homology

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Abstract

We present a robust zk-SNARK construction enabling a prover to convince a verifier that a hidden graph G is a spine of a public cubical handlebody $N \subset \mathbb{R}^3$. By leveraging Nielsen's theorem and fixed meridian disks, we reduce the non-abelian homotopy equivalence problem to an integer homology isomorphism. The arithmetic circuit (R1CS) enforces topological inclusion, connectivity via a rooted directed spanning arborescence, and homological unimodularity. We correct previous computational fallacies by implementing a rigorous integer intersection matrix calculation over undirected edges with a fixed orientation, and a validated shifted-representation for the inverse matrix. Our construction achieves perfect completeness and computational knowledge soundness under standard assumptions.

1 Introduction

Zero-knowledge succinct non-interactive arguments of knowledge (zk-SNARKs) [1, 2] enable a prover to verify complex statements without revealing the witness. We apply this paradigm to a topological problem: proving that a hidden graph G embedded in a public cubical handlebody N is a *spine* of N , i.e., N strong deformation retracts onto G . By Whitehead's theorem, since N and G are $K(\pi, 1)$ spaces, G is a spine if and only if $\pi_1(G) \cong \pi_1(N)$. Nielsen's theorem reduces this to checking an isomorphism on integer homology $H_1(\cdot; \mathbb{Z})$, which is amenable to algebraic circuit constraints.

2 Topological and Geometric Preliminaries

2.1 Cubical Complex and Inclusion

Let $\mathcal{C}$ be the 1-skeleton of an $M \times M \times M$ grid. The handlebody N is a connected union of 3-cells (cubes) of this grid. The verifier precomputes two public indicator vectors:

$$\text{inN_v}(v) \in \{0, 1\}, \forall v \in V(\mathcal{C}); \quad \text{inN_e}(e) \in \{0, 1\}, \forall e \in E(\mathcal{C}).$$

The hidden graph $G = (V_G, E_G)$ must be a subgraph of $\mathcal{C}$ completely contained in N .

2.2 Spines and Homology

A subspace $G \subseteq N$ is a *spine* if N strong deformation retracts onto G . For a handlebody and a graph, higher homotopy groups vanish; therefore G is a spine iff $\beta_1(G) = g$ and the inclusion $\iota: G \hookrightarrow N$ induces an isomorphism $H_1(G; \mathbb{Z}) \rightarrow H_1(N; \mathbb{Z})$.

2.3 Meridian Disks and Intersection Numbers

The verifier fixes g meridian disks $\{D_j\}_{j=1}^g$ in N , constructed as dual 2-cells of the cubical complex; they form a basis of $H_2(N, \partial N)$. For each undirected edge $e \in E(\mathcal{C})$ we fix an arbitrary reference orientation (e.g., from the endpoint with smaller index to the larger one). The *algebraic intersection* of e with D_j with respect to this orientation is a public constant $\lambda_{e,j} \in \{-1, 0, 1\}$.

3 The Arithmetic Circuit (R1CS)

3.1 Witness Variables

The prover provides:

- $v_A(v) \in \{0, 1\}$: vertex v is in G ,
- $g_e \in \{0, 1\}$: undirected edge e is in G ,
- $t_{\vec{e}} \in \{0, 1\}$: directed edge $\vec{e}$ belongs to the spanning arborescence T ,
- $h(v) \in \mathbb{F}_p$: height of v in T ,
- $b_{v,k} \in \{0, 1\}$: bits of $h(v)$ for range proofs,
- $x_{e,i} \in \{-1, 0, 1\}$: flow of the i -th basis cycle on the undirected edge e (relative to the fixed orientation),
- $\widehat{W}_{i,j} \in \mathbb{F}_p$: shifted inverse matrix entries.

3.2 Inclusion and Boolean Constraints

$$v_A(v)(1 - v_A(v)) = 0,$$

$$g_e(1 - g_e) = 0,$$

$$v_A(v)(1 - \text{inN}_{\mathbf{v}}(v)) = 0, \tag{1}$$

$$g_e(1 - \text{inN}_{\mathbf{e}}(e)) = 0, \tag{2}$$

$$g_e(1 - v_A(\text{end}_1(e))) = 0, \tag{3}$$

$$g_e(1 - v_A(\text{end}_2(e))) = 0. \tag{4}$$

3.3 Rooted Spanning Arborescence

To prove connectivity and $\beta_1(G) = g$, the prover selects a root $v_0 \in V(\mathcal{C}) \cap N$ and commits to it as part of the witness, enforcing $v_A(v_0) = 1$ and $\text{inN}_{\mathbf{v}}(v_0) = 1$. The directed tree T must satisfy:

$$t_{\vec{e}}(1 - t_{\vec{e}}) = 0,$$

$$(t_{\vec{e}} + t_{-\vec{e}})(1 - g_e) = 0, \tag{5}$$

$$t_{\vec{e}}t_{-\vec{e}} = 0, \tag{6}$$

$$\sum_{\vec{e}: \text{head}(\vec{e})=v} t_{\vec{e}} = v_A(v) \quad (v \neq v_0), \tag{7}$$

$$\sum_{\vec{e}: \text{head}(\vec{e})=v_0} t_{\vec{e}} = 0.$$

Height constraints prevent cycles (with $K = \lceil \log_2 |\mathcal{C}| \rceil$):

$$\begin{aligned} h(v_0) &= 0, \\ b_{v,k}(1 - b_{v,k}) &= 0, \\ h(v) - \sum_{k=0}^{K-1} 2^k b_{v,k} &= 0, \end{aligned} \tag{8}$$

$$t_{\vec{e}}(h(\text{head}(\vec{e})) - h(\text{tail}(\vec{e})) - 1) = 0. \tag{9}$$

These guarantee T is an acyclic arborescence spanning exactly the active vertices, so $|E_T| = |V_G| - 1$. The rank condition is enforced by

$$\sum_e g_e - \sum_v v_A(v) + 1 = g. \tag{10}$$

3.4 Cycle Flows and Homology Matrix

For each basis cycle $i = 1, \dots, g$, the flow $x_{e,i} \in \{-1, 0, 1\}$ is constrained by:

$$x_{e,i}(x_{e,i} - 1)(x_{e,i} + 1) = 0, \tag{11}$$

$$x_{e,i}(1 - g_e) = 0, \tag{12}$$

$$\sum_{e \in E(\mathcal{C})} s_{v,e} x_{e,i} = 0 \quad \forall v, \tag{13}$$

where $s_{v,e} = +1$ if the fixed orientation of e points away from v , -1 if towards v , and 0 otherwise. (As noted in §3.1, the degree-3 polynomial (11) is reduced to degree-2 via $y = x^2$ and $y(1 - y) = 0$.)

The homological intersection matrix $M \in M_{g \times g}(\mathbb{Z})$ is

$$M_{i,j} = \sum_{e \in E(\mathcal{C})} x_{e,i} \lambda_{e,j}. \tag{14}$$

Because we sum over undirected edges exactly once, M correctly represents ι_* without the artificial factor 2.

3.5 Unimodularity over Integers

The map ι_* is an isomorphism iff $\det M = \pm 1$. The prover supplies an inverse matrix W . To prevent finite-field wrap-around, we use a shifted representation and the Hadamard bound. Set $L = |E(\mathcal{C})|$; then $|M_{i,j}| \leq L$. By Cramer's rule,

$$B_W = (g - 1)^{\frac{g-1}{2}} L^{g-1} \tag{15}$$

bounds every entry of W (with $B_W = 1$ when $g = 1$). The prover commits to $\widehat{W}_{i,j} = W_{i,j} + B_W$ and proves

$$\widehat{W}_{i,j} \in [0, 2B_W] \quad (\text{via bit-decomposition}). \tag{16}$$

The circuit computes $W_{i,j} = \widehat{W}_{i,j} - B_W$ and checks

$$M \cdot W = I \pmod{p}, \tag{17}$$

choosing the field prime $p > 2gLB_W$ to guarantee that the modular identity lifts to exact integer equality $MW = I$.

4 Security Analysis

Theorem 4.1 (Perfect Completeness). An honest prover with a spine G can always construct a satisfying witness.

Proof. A spine G is connected, has $\beta_1(G) = g$, and the inclusion induces a unimodular intersection matrix M . The prover computes a spanning tree T (e.g., via DFS) and extracts g fundamental cycles, obtaining integer flows $x_{e,i} \in \{-1, 0, 1\}$. The inverse W is integral, and its shifted representation satisfies all range constraints. $\square$

Theorem 4.2 (Computational Knowledge Soundness). Under the Knowledge-of-Exponent assumption for the underlying SNARK, any accepted proof extracts a valid spine.

Proof. The inclusion masks (1)–(4) force $G \subseteq N$. The arborescence constraints together with the Euler equation (10) guarantee G is connected and $\beta_1(G) = g$. The cycle constraints (11)–(13) ensure that the flows represent valid 1-cycles. The matrix M defined by (14) is therefore the matrix of ι_* relative to the meridian basis. The shift and Hadamard bounds prove M is unimodular over $\mathbb{Z}$. Hence ι_* is an isomorphism. By Nielsen’s theorem $\pi_1(G) \cong \pi_1(N)$, and by Whitehead’s theorem (since both spaces are $K(\pi, 1)$) G is a deformation retract, i.e., a spine. $\square$

Zero-Knowledge. The SNARK simulator produces proofs indistinguishable from real ones, so no geometric information of G is revealed.

5 Practicality and Complexity

The circuit size scales as $O(|E(\mathcal{C})| \cdot g + |V(\mathcal{C})| \log |\mathcal{C}|)$. For $g = 5$ and $|\mathcal{C}| \approx 10^3$, $B_W \approx 1.6 \cdot 10^{13}$, easily handled by a 254-bit field. For larger parameters, multi-precision techniques are needed but remain asymptotically feasible.

References

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